Elective IV

UEI744 ROBUST AND REAL TIME CONTROL SYSTEMS

L	T	P	Cr.
3	0	0	3.0

Course Objectives: The objective of the course is to provide basic understanding of real time control systems and to provide basic introduction to modern robust control theory techniques for large-scale uncertain multivariable systems: stability and performance; computer-aided tools for both system analysis and controller design.

Classical Control Review: Internal Stability, stationary behaviour of the feedback loop, requirements for feedback systems: performance, robust stability, robust performance.

Norms of Systems and Performance: Vector norms, signal norms, system norms, calculation of operator norms, specification for feedback systems, performance limitations. Multi-input multi- output (MIMO) systems: Singular values, structured singular value.

Feedback interconnection, stability, Uncertainty and robustness: Well-pawedness; Internal stability; Coprime factorization and stabilizing controllers, uncertainty representations; Uncertain polynomials; Boundary crossing theorem; Kharitonov's result; Edge theorem; Stability of polytope of polynomials; Sensitivity and complementary sensitivity; Linear fractional transformation (LFT), Robust stability.

H2 optimal control: LQ controllers, Kalman filter, LQG controllers, characterization of H2 optimal controllers.

Real-time Implementation: Real-time Systems Implementation Techniques, Analog Controller realization, circuit implementation, microprocessor control of control system, Digital signal processor-based implementation, effect of quantization, Time delays in control system realization. Concurrent Programming. Static and Dynamic allocation Algorithms: First cum first serve, shortest job first, Allocation with priority.

Course Learning Outcomes (CLO):

Student will be able to:

1. Analyze multivariable linear control system.
2. Analyze stability and robustness of control system
3. Characterize and formulate H2 Optimal Control problem
4. Demonstrate the knowledge of Implementation Techniques of real-time systems

Text Books:

1. Jeffrey B. Burl, Linear optimal control, H2 and H∞ methods, Addison-Wesley, 1999.
2. K. Zhou, J. Doyle, and K. Glover. Robust and Optimal Control. Prentice Hall, Englewood Cliffs, New Jersey, 1996.
3. U. Mackenroth, Robust Control Systems: Theory and case studies. Springer, 2004.

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

4. D.W. Gu, P. Hr. Petkov and M.M. Konstantinov, Robust Control Design with MATLAB. Springer 2005.
5. Tian Seng Ng , Real Time Control Engineering: Systems And Automation; Springer, 2016

Reference Books:

1. Green M & Limebeer D.J., Linear robust control; Courier Corporation; 2012.
2. Zhou, Kemin, and John Comstock Doyle. Essentials of robust control. Vol. 104. Upper Saddle River, NJ: Prentice hall, 1998.
3. Peng Zhang, Industrial Control Technology: A Handbook for Engineers and Researchers; William Andrel, 2007.

Evaluation Scheme:

S.No.	**Evaluation Elements**	**Weightage (%)**
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30